Day 4 Task Workflow — Friday, Feb 28, 2026

> **Sprint:** 1 (Foundation) | **Phase:** Redis Integration & Frontend Shell
> **Owner:** Lead Architect + Full-Stack Dev | **Status:** ⬤ Ready to Start
> **Goal:** Redis pub/sub operational, Next.js frontend shell with sidebar navigation, all 10 module stub pages

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Day 4 Objective

Complete the real-time communication layer and frontend navigation so that by end of day:

- Redis connection established and pub/sub test verified
- Next.js frontend shell: sidebar navigation with icon-only design
- All 10 module pages created as stub pages
- Navigation working with active state highlighting
- TopBar with threat level indicator, notifications, user menu
- StatusBar with system status footer
- Dark theme War Room design language applied

Scope Adherence Checklist

> **STRICT RULE:** Every task below MUST adhere to the master documentation specifications. No features outside the defined scope.

Requirement	Source Document	Section
Redis 7 for pub/sub	GLOBAL_CONTEXT.md	§2 (Technology Stack)
Next.js 16 App Router	GLOBAL_CONTEXT.md	§2
TypeScript strict mode	GLOBAL_CONTEXT.md	§9 (Rule #9)
Dark theme only	GLOBAL_CONTEXT.md	§9 (Rule #10)
Glassmorphism UI	GLOBAL_CONTEXT.md	§9 (Rule #10)
Cyan accents (#00f0ff)	GLOBAL_CONTEXT.md	§9
10 modules only	GLOBAL_CONTEXT.md	§5
Sidebar icon-only design	MASTER_DOC_PART5	§2.1 (Sidebar.tsx)
War Room / Intel Agency vibe	GLOBAL_CONTEXT.md	§9 (Rule #10)

Task Breakdown

TASK 1 — Redis Connection & Pub/Sub Test 🔨

Time Est: 30 min | **Priority:** ⬤ Critical

Set up Redis connection manager and verify pub/sub functionality.

1.1 Redis Client (`backend/app/redis.py`)

Component	Purpose	Implementation
`RedisManager`	Connection manager	`redis.asyncio` client
`connect()`	Establish connection	Pool with health check
`publish()`	Publish to channel	`await redis.publish(channel, message)`
`subscribe()`	Subscribe to channel	`await redis.subscribe(channel)`
`get/set`	Cache operations	For LLM response caching

Key Implementation Points:

- Use `redis.asyncio` for async operations (matches FastAPI async pattern)
- Connection URL from `settings.REDIS_URL`
- Connection pooling with `max_connections=20`
- Health check ping on startup
- Graceful shutdown on app termination

Channels to Define:

Channel	Purpose	Publisher	Subscriber
`flows:live`	Real-time flow data	Capture Engine	ML Worker, WebSocket
`alerts:live`	New/updated alerts	Alert Engine	WebSocket
`system:status`	System metrics	Backend	WebSocket

Verification:

```
cd backend && python -c "
import asyncio
import redis.asyncio as redis

async def test():
    r = redis.from_url('redis://localhost:6379')
    await r.set('test_key', 'hello')
    val = await r.get('test_key')
    print('[OK] Redis get/set:', val.decode())
    await r.delete('test_key')
    await r.close()

asyncio.run(test())
"
```

Expected Output:

```
[OK] Redis get/set: hello
```

TASK 2 — Redis Integration in FastAPI 🔨

Time Est: 20 min | **Priority:** ⬛ Critical

Integrate Redis into the FastAPI application lifecycle.

2.1 Update Main App (`backend/app/main.py`)

Change	Purpose
Add Redis to lifespan	Connect on startup, disconnect on shutdown
Add Redis dependency	`get_redis()` for endpoint access
Health check integration	Include Redis status in `/system/health`

Lifespan Pattern:

```python
@asynccontextmanager
async def lifespan(app: FastAPI):
    # Startup
    redis_manager = RedisManager(settings.REDIS_URL)
    await redis_manager.connect()
    app.state.redis = redis_manager

    yield

    # Shutdown
    await redis_manager.disconnect()
```

Verification:

```
cd backend && python -c "
from app.main import app
print('[OK] App has redis_manager:', hasattr(app.state, 'redis_manager'))
"
```

TASK 3 — Next.js Frontend Shell: Root Layout 🔨

Time Est: 45 min | **Priority:** ⬛ Critical

Create the root layout with sidebar, topbar, and statusbar.

3.1 Root Layout (`frontend/app/layout.tsx`)

Structure (per MASTER_DOC_PART5 §2.1):

```
┌──────────────────────────────────────┐
│ Sidebar │ TopBar                     │
│ (icon)  ├────────────────────────────┤
│         │ Main Content (scrollable)  │
│         ├────────────────────────────┤
│         │ StatusBar                  │
└──────────────────────────────────────┘
```

Key Elements:

- Import `globals.css` for design system
- Set `lang="en"` on `<html>`
- Apply dark theme class to `<body>`
- Grid layout: sidebar (64px) + main content
- Font imports: JetBrains Mono, Inter

Verification:

```
cd frontend && npm run build 2>&1 | grep -E "(error|Error)" || echo "[OK] Build successful"
```

TASK 4 — Sidebar Navigation Component 🔨

Time Est: 60 min | **Priority:** 🔴 Critical

Create icon-only sidebar with navigation to all 10 modules.

4.1 Sidebar Component (`frontend/components/layout/Sidebar.tsx`)

Design Specifications (per GLOBAL_CONTEXT.md §9):

- Width: 64px (icon-only)
- Background: `#111118` (panel color)
- Border: `rgba(255, 255, 255, 0.06)`
- Active state: cyan glow (`#00f0ff`)
- Icons: Lucide React

Navigation Items (10 modules):

Icon	Label	Route	Priority
`Shield`	War Room	`/war-room`	P0

Icon	Label	Route	Priority
`Search`	Threat Hunt	`/hunt`	P0
`Globe`	Intel Hub	`/intel`	P0
`Network`	Network Flow	`/network`	P0
`Bot`	AI Analyst	`/ai-analyst`	P0
`Bell`	Alert Console	`/alerts`	P1
`FileSearch`	Forensics Lab	`/forensics`	P1
`Brain`	ML Operations	`/ml-ops`	P1
`FileText`	Reports	`/reports`	P1
`Settings`	Administration	`/admin`	P2

Active State Logic:

- Use `usePathname()` from `next/navigation`
- Match current route to nav item
- Apply `.active` class with cyan glow effect

Verification:

```
cd frontend && npm run build 2>&1 | grep -E "(error|Error)" || echo "[OK] Sidebar
component builds"
```

TASK 5 — TopBar Component 🔨

Time Est: 30 min | **Priority:** ⬤ Critical

Create top bar with threat level, notifications, and user menu.

5.1 TopBar Component (`frontend/components/layout/TopBar.tsx`)

Elements:

Element	Description	Style
Threat Level Indicator	DEFCON-style gauge (1-5)	Color-coded: red→yellow→green
Notification Bell	Alert count badge	Cyan accent
User Menu	Avatar + dropdown	Glass panel
Search Icon	Global search trigger	Future implementation

Threat Levels:

Level	Color	Meaning
1 - CRITICAL	`#ef4444`	Active attack detected
2 - HIGH	`#f97316`	Multiple anomalies
3 - ELEVATED	`#eab308`	Suspicious activity
4 - GUARDED	`#22c55e`	Normal operations
5 - LOW	`#00f0ff`	All clear

Verification:

```
cd frontend && npm run build 2>&1 | grep -E "(error|Error)" || echo "[OK] TopBar
component builds"
```

TASK 6 — StatusBar Component 🔨

Time Est: 15 min | **Priority:** 🟡 Medium

Create system status footer bar.

6.1 StatusBar Component (`frontend/components/layout/StatusBar.tsx`)

Elements:

Element	Description
Connection Status	WebSocket connected/disconnected indicator
Capture Status	Engine running/stopped
Last Update	Timestamp of last data refresh
Version	App version number

Style:

- Height: 32px
- Background: `#0a0a0f` (base)
- Font: JetBrains Mono (monospace)
- Text color: `#64748b` (muted)

Verification:

```
cd frontend && npm run build 2>&1 | grep -E "(error|Error)" || echo "[OK]
StatusBar component builds"
```

TASK 7 — All 10 Module Stub Pages 🔨

Time Est: 30 min | **Priority:** 🔴 Critical

Create stub pages for all 10 modules with consistent structure.

7.1 Page Structure

Each page should have:

- Metadata export (title, description)
- GlassPanel wrapper
- Page title with module name
- "Coming Soon" placeholder content

Pages to Create:

#	File	Route	Title
1	`app/war-room/page.tsx`	`/war-room`	War Room
2	`app/hunt/page.tsx`	`/hunt`	Threat Hunt
3	`app/intel/page.tsx`	`/intel`	Intel Hub
4	`app/network/page.tsx`	`/network`	Network Flow
5	`app/ai-analyst/page.tsx`	`/ai-analyst`	AI Analyst
6	`app/alerts/page.tsx`	`/alerts`	Alert Console
7	`app/forensics/page.tsx`	`/forensics`	Forensics Lab
8	`app/ml-ops/page.tsx`	`/ml-ops`	ML Operations
9	`app/reports/page.tsx`	`/reports`	Reports
10	`app/admin/page.tsx`	`/admin`	Administration

Stub Page Template:

```
import { Metadata } from "next";
import { GlassPanel } from "@/components/shared/GlassPanel";

export const metadata: Metadata = {
  title: "Module Name — ThreatMatrix AI",
  description: "Module description",
};

export default function ModulePage() {
  return (
    <div className="page-container">
      <GlassPanel>
        <h1 className="page-title">Module Name</h1>
```

```
          <p className="text-muted">Module coming soon...</p>
        </GlassPanel>
      </div>
    );
  }
```

Verification:

```
cd frontend && ls -la app/*/page.tsx | wc -l
# Expected: 10 (or 11 including root page.tsx)
```

TASK 8 — Root Page Redirect 🔨

Time Est: 5 min | **Priority:** 🟢 Medium

Update root page to redirect to `/war-room`.

8.1 Update `frontend/app/page.tsx`

```
import { redirect } from "next/navigation";

export default function Home() {
  redirect("/war-room");
}
```

Verification:

```
cd frontend && npm run build 2>&1 | grep -E "(error|Error)" || echo "[OK] Root
redirect builds"
```

TASK 9 — Design System CSS Verification 🔨

Time Est: 15 min | **Priority:** 🟢 Medium

Verify globals.css has all required design tokens.

9.1 Required CSS Variables (per GLOBAL_CONTEXT.md §9)

Variable	Value	Purpose
`--bg-primary`	`#0a0a0f`	Main background
`--bg-secondary`	`#111118`	Panel background

Variable	Value	Purpose
`--bg-tertiary`	`#1a1a24`	Card background
`--cyan`	`#00f0ff`	Primary accent
`--critical`	`#ef4444`	Critical alerts
`--high`	`#f97316`	High severity
`--safe`	`#22c55e`	Safe/normal
`--font-data`	`JetBrains Mono`	Data/metrics
`--font-ui`	`Inter`	UI labels

Verification:

```
cd frontend && grep -E "(--bg-primary|--cyan|--critical|--font-data)"
styles/globals.css | head -5
```

TASK 10 — Full Stack Verification 🔨

Time Est: 15 min | **Priority:** 🔴 Critical

Verify all components work together.

10.1 Verification Checklist

Check	Command	Expected
Frontend builds	`cd frontend && npm run build`	No errors
All pages exist	`ls app/*/page.tsx`	10+ pages
Sidebar renders	Visual check at `localhost:3000`	Icon navigation visible
Navigation works	Click sidebar icons	Routes change, active state updates
Dark theme	Visual check	Dark background, cyan accents
Backend starts	`cd backend && uvicorn app.main:app`	No errors
Redis connects	Check `/api/v1/system/health`	Redis status: connected

Files Created/Modified Today

```
threatmatrix-ai/
├── backend/
│   ├── app/
│   │   ├── redis.py                    🔨 Task 1
```

```
│   │   └── main.py                  🔨 Task 2 (update)
├── frontend/
│   ├── app/
│   │   ├── layout.tsx               🔨 Task 3 (verify/update)
│   │   ├── page.tsx                 🔨 Task 8 (update)
│   │   ├── war-room/page.tsx        🔨 Task 7
│   │   ├── hunt/page.tsx            🔨 Task 7
│   │   ├── intel/page.tsx           🔨 Task 7
│   │   ├── network/page.tsx         🔨 Task 7
│   │   ├── ai-analyst/page.tsx      🔨 Task 7
│   │   ├── alerts/page.tsx          🔨 Task 7
│   │   ├── forensics/page.tsx       🔨 Task 7
│   │   ├── ml-ops/page.tsx          🔨 Task 7
│   │   ├── reports/page.tsx         🔨 Task 7
│   │   └── admin/page.tsx           🔨 Task 7
│   └── components/
│       └── layout/
│           ├── Sidebar.tsx          🔨 Task 4
│           ├── TopBar.tsx           🔨 Task 5
│           └── StatusBar.tsx        🔨 Task 6
```

Verification Checklist

> **Every item below MUST be verified before marking task complete.**

#	Verification	Command	Expected Result
1	Redis connection	Test script	`[OK] Redis get/set: hello`
2	Frontend builds	`npm run build`	No errors
3	All 10 pages exist	`ls app/*/page.tsx`	10+ pages
4	Sidebar renders	Visual check	Icon navigation visible
5	Navigation works	Click icons	Routes change, active state
6	Dark theme	Visual check	Dark bg, cyan accents
7	TopBar renders	Visual check	Threat level, notifications
8	StatusBar renders	Visual check	Connection status visible
9	Root redirect	Visit `/`	Redirects to `/war-room`
10	Design tokens	grep globals.css	All required vars present

Scope Adherence Verification

Requirement	Source	Verification
Redis 7 for pub/sub	GLOBAL_CONTEXT.md §2	Check redis.asyncio usage

Requirement	Source	Verification
Next.js 16 App Router	GLOBAL_CONTEXT.md §2	Check app/ directory structure
TypeScript strict mode	GLOBAL_CONTEXT.md §9	Check tsconfig.json strict: true
Dark theme only	GLOBAL_CONTEXT.md §9	Visual verification
Glassmorphism UI	GLOBAL_CONTEXT.md §9	Check GlassPanel component
Cyan accents (#00f0ff)	GLOBAL_CONTEXT.md §9	Check CSS variables
10 modules only	GLOBAL_CONTEXT.md §5	Count page.tsx files
Sidebar icon-only	MASTER_DOC_PART5 §2.1	Check Sidebar.tsx width: 64px
War Room vibe	GLOBAL_CONTEXT.md §9	Visual verification

Blockers

Blocker	Severity	Mitigation	Status
Docker not running	◍ Medium	Redis can be tested via remote or wait for Docker fix	Mitigated
No live data yet	◉ Low	Use mock/placeholder data for UI verification	Expected

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Tomorrow's Preview (Day 5)

- Design system components: GlassPanel, StatusBadge, MetricCard, DataTable
- Docker stack verification (if Docker fixed)
- API docs accessible at `/docs`
- Frontend connecting to backend API

Task workflow for Day 4 — ThreatMatrix AI Sprint 1